

Metric-Semantic 6D Pose Estimation of PC Back-Panel Components

Malika T M Yashita Jaiswal

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Abstract

We present a multi-view geometric pipeline for estimating the 6-DoF poses of PC back-panel components from 16 posed RGB images with known camera intrinsics. The method triangulates bounding box corners across multiple views to recover a sparse 3D point cloud, followed by plane fitting to enforce the planar structure of the panel. An oriented bounding box (OBB) is constructed using the estimated plane normal and principal directions. The 3D center is further refined via multi-view reprojection optimization. The approach is robust to low-texture regions and produces physically consistent pose estimates.

1 Problem Description

The goal is to estimate the 6D pose of components such as power sockets and ethernet ports on a PC back panel using:

- 16 RGB images
- Known camera-to-world poses
- Camera intrinsics

Each object is represented as an oriented bounding box (OBB) defined by a 3D center, extents, and a rotation matrix.

2 Methodology

2.1 Data Preparation

Camera intrinsics ($f_x = 1477.0$, $f_y = 1480.4$, $c_x = 1298.3$, $c_y = 686.8$) are used to construct the calibration matrix K . Camera poses are provided as 4×4 camera-to-world matrices and inverted to obtain world-to-camera transformations.

Manual annotations provide four bounding box corners per object in multiple frames. These are converted into consistent corner correspondences.

2.2 Multi-View Triangulation

For each component, the four bounding box corners are triangulated across all pairs of annotated frames. Given projection matrices

$$P = K[R \mid t],$$

3D points are recovered using linear triangulation.

Each triangulated point is filtered using:

- Cheirality constraint (positive depth in both cameras)

- Reprojection error threshold of 50 pixels

Outliers are removed using Median Absolute Deviation (MAD) with a threshold of 3.0. This yields approximately 200–300 valid 3D points per component.

2.3 Plane Fitting

Since all components lie on a planar back panel, a plane is fitted to the point cloud using Singular Value Decomposition (SVD). The plane normal is taken as the singular vector corresponding to the smallest singular value.

To ensure consistency, the normal direction is oriented towards the camera. All points are projected onto the estimated plane, reducing depth noise and enforcing geometric structure.

2.4 6D Pose Construction

The oriented bounding box is constructed as follows:

- **Center:** mean of the plane-projected points
- **Axes:**
 - One axis aligned with the plane normal
 - One axis fixed as the global horizontal direction
 - Third axis obtained via cross product
- **Extents:** computed using PCA spread of the point cloud

The depth extent is constrained using a fixed physical prior of 6 mm, corresponding to the approximate protrusion of connectors from the panel. Maximum extents are also capped using known physical limits for each component.

2.5 Reprojection Refinement

The estimated 3D center is refined by minimizing reprojection error across all annotated frames:

$$\sum_i \left\| \mathbf{x}_i^{proj} - \mathbf{x}_i^{obs} \right\|^2$$

Optimization is performed using the Nelder-Mead method with a maximum of 5000 iterations. This step reduces multi-view inconsistency and improves localization accuracy, typically by a few millimeters.

3 Results

The pipeline produces stable pose estimates for all components. Key observations:

- Approximately 200–300 valid triangulated points per object
- Plane fitting significantly stabilizes orientation
- Reprojection refinement reduces center error

The estimated extents are physically consistent with real connector dimensions, and reprojection errors remain low across all views.

4 Discussion

The method is effective for low-texture planar scenes where feature-based methods fail. Using structured corner correspondences and enforcing planar constraints improves robustness.

Limitations include sensitivity to annotation accuracy and dependence on sufficient view-point variation for reliable triangulation.

5 Conclusion

We presented a robust pipeline for 6D pose estimation using multi-view geometry and planar constraints. The combination of corner-based triangulation, plane fitting, and reprojection refinement produces accurate and physically meaningful results from a sparse set of images.